

1. Administrative & Study Identification

Full Study Title: Substudy 02C: Safety and Efficacy of Pembrolizumab in Combination With Investigational Agents or Pembrolizumab Alone in Participants With Stage III Melanoma Who Are Candidates for Neoadjuvant Therapy (MK-3475-02C/KEYMAKER-U02)
Short Title/Acronym: KEYMAKER-U02 **Substudy 02C Protocol Number:** MK-3475-02C **NCT Number:** NCT04303169 **Sponsor Name:** Merck
Investigational Product: Pembrolizumab (KEYTRUDA®) alone or in combination with investigational agents (Vibostolimab, Gebasaxturev, MK-4830, Favezelimab, or all-trans retinoic acid [ATRA] / tretinoin) **Phase of Development:** Phase 1/Phase 2 **Medical Monitor:** Clinical Operations Lead, Merck

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability (including adverse event incidence and discontinuations due to AEs), and the pathological complete response (pCR) rate of investigational treatment arms when used as neoadjuvant therapy in participants with Stage III melanoma.
Secondary Objectives: To evaluate the near pathological complete response (near pCR) rate, pathological partial response (pPR) rate, and recurrence-free survival (RFS).
2.2 Background & Rationale To address the need for improved outcomes in high-risk, resectable Stage III melanoma. While adjuvant therapies exist, neoadjuvant approaches (treatment administered prior to surgery) aim to reduce tumor burden, facilitate surgery, and potentially improve long-term recurrence-free survival. This umbrella substudy evaluates if adding novel investigational agents to pembrolizumab is superior to the current treatment option of pembrolizumab monotherapy/historical control.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Open-Label Rolling-Arm Umbrella Platform Design) **Control Type:** Active Comparator (Pembrolizumab monotherapy arm) **Blinding:** Open-label
Randomization: Randomized (participants are randomly assigned to open arms)
3.2 Planned Enrollment & Duration Target \$N\$: 146 participants
Number of Sites: Multicenter (Global) **Trial Status:** COMPLETED

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed clinically detectable and resectable Stage IIIB, IIIC, or IIID melanoma that is amenable to surgical resection. 2. Untreated for Stage IIIB, IIIC, or IIID melanoma (prior surgical resection or radiotherapy to the primary melanoma is allowed). 3. Adequate organ function and provision of a baseline tumor biopsy. 4. Resolution of toxic effect(s) of the most recent prior therapy to Grade 1 or less (except alopecia). 5. Contraception/Abstinence: * Male participants receiving gebasaxturev must be abstinent or use contraception during intervention and for ≥ 120 days post-dose; those receiving ATRA for ≥ 7 days post-dose. * Female participants must not be pregnant/breastfeeding and must be abstinent or use highly effective contraception during intervention and for ≥ 120 days after the last dose of pembrolizumab, vibostolimab, gebasaxturev, MK-4830, or favezelimab + pembrolizumab, or ≥ 30 days after the last dose of ATRA.

4.2 Exclusion Criteria 1. Diagnosis of ocular or mucosal melanoma (or has only mucosal lesions). 2. Active autoimmune disease requiring systemic treatment in the past 2 years, or receiving immunosuppressive therapy within 7 days before the first dose. 3. Known additional malignancy progressing or requiring active treatment within the past 2 years. 4. Known central nervous system (CNS) metastases and/or carcinomatous meningitis. 5. Known hypersensitivity, including previous clinically significant reaction to another monoclonal antibody (mAb). 6. Active infection requiring systemic therapy, known history of HIV, Hepatitis B, active tuberculosis (TB), or history of (noninfectious) pneumonitis. 7. Prior systemic anticancer therapy within 4 weeks, prior radiotherapy within 2 weeks, or major surgery < 3 weeks prior to randomization/first dose. 8. Received a live or live-attenuated vaccine within 30 days before the first dose. 9. Allogeneic tissue/solid organ transplant. 10. Not naïve to Talimogene laherparepvec (TVEC) and other oncolytic viruses.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Neoadjuvant Phase:* Participants are assigned to one of the following arms: * **Arm 1:** Pembrolizumab + Vibostolimab * **Arm 2:** Pembrolizumab + Gebasaxturev * **Arm 3:** Pembrolizumab * **Arm 4:** Pembrolizumab + MK-4830 * **Arm 5:** Favezelimab + Pembrolizumab * **Arm 6:** Pembrolizumab + all-trans retinoic acid (ATRA) * *Adjuvant Phase (Post-Surgery):* Pembrolizumab 400 mg Q6W for up to 8 cycles. **Route of Administration:** Intravenous (IV) for Pembrolizumab, Vibostolimab, MK-4830, Favezelimab; Intratumoral (IT) for Gebasaxturev; Oral (PO) for ATRA. **Duration of Treatment:** Up to approximately 1 year total (neoadjuvant plus adjuvant).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care supportive therapies and postoperative management medications. **Prohibited:** Concurrent immunosuppressive medications, live virus vaccines (within 30 days of first dose), and other investigational antineoplastic therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Medical History, Baseline Tumor Biopsy, Eligibility Confirmation
2	0 to 6	Administration of Pembrolizumab Investigational Agent(s) prior to surgery
3	6 to 12	Surgical resection of tumor, Pathological response assessment
4	12 onwards	Initiation of Adjuvant Pembrolizumab (400 mg Q6W for up to 8 cycles)
5	Up to 1 Year	Final safety/efficacy assessments, survival follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Percentage of participants who experience an adverse event (AE). 2. Percentage of participants who discontinue study treatment due to an AE. 3. **Pathological Complete Response (pCR) Rate:** Defined as the proportion of participants with complete absence of viable tumor in the treated tumor bed. **Measurement Method:** Assessments for pCR are according to Response Evaluation Criteria in Solid Tumors 1.1 (RECIST 1.1) by central review of the pathology results. RECIST 1.1 has been modified for this study to include a maximum of 10 target lesions and a maximum of 5 target lesions per organ. Clinical safety monitoring evaluates AEs and discontinuations.

7.2 Secondary Outcomes 1. Near pathological complete response (near pCR) rate. 2. Pathological partial response (pPR) rate. 3. Recurrence-free survival (RFS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient assessments, lab tests, and physical exams (tracking any untoward medical occurrence, including Grade 3 or 4 Treatment-Related AEs). **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External/internal safety monitoring overseen by the sponsor for the KEYMAKER-U02 umbrella protocol.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes (pCR). Safety Set for all participants who receive at least 1 dose of study treatment for AE/SAE evaluations. **Sample Size Justification:** Adaptive umbrella design allocating participants across multiple sub-arms (Target N=146) to detect early safety profiles and pathological response signals. **Handling of Missing Data:** Patients with missing pathological response data are considered non-responders for pCR. Survival endpoints (RFS) utilize standard Kaplan-Meier estimates and censoring.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to ensure protocol compliance and data integrity throughout the neoadjuvant and adjuvant periods. **Audit Trail:** Maintained via validated Electronic Data Capture (EDC) systems with full audit trails, eCRF tracking, and central pathology data clarification.

11. Study Identifiers & Governance

Primary ID: MK-3475-02C **Registry Number:** NCT04303169 **Sponsor/Lead Organization:** Merck **Study Title:** KEYMAKER-U02 Substudy 02C **Official Regulatory Title:** A Phase 1/2 Open-Label Rolling-Arm Umbrella Platform Design of Investigational Agents With or Without Pembrolizumab or Pembrolizumab Alone in Participants With Melanoma (KEYMAKER-U02): Substudy 02C

12. Clinical Framework (Melanoma Specific)

Primary Condition: Malignant Melanoma (Stage IIIB, IIIC, IIID) **Semantic Type:** Neoplastic Process **Diagnosis Threshold:** Histologically or cytologically confirmed, clinically detectable and resectable Stage III melanoma. **Medical Methodology:** Neoadjuvant Immunotherapy $\rightarrow$ Surgical Resection $\rightarrow$ Adjuvant Immunotherapy **Optimization Target:** Pathological Complete Response (pCR) and Recurrence-Free Survival (RFS)

13. Study Procedures & Methodology

Management Pathway: Standard neoadjuvant/adjuvant clinical pathway for resectable melanoma. **Education Protocol (HCP):** Training on investigational agent administration (especially

intratumoral injections for Gebasaxturev) and immune-mediated AE management. **Education Protocol (Patient):** Informed consent, strict contraception/abstinence rules based on assigned study arm, treatment schedule expectations, and post-surgical recovery/ adjuvant monitoring. **Quality Audit Mechanism:** Standard clinical trial monitoring and rigorous central/local pathology review for accurate pCR/pPR determination using modified RECIST 1.1 criteria.

Total Duration: Up to 1 year of active treatment, followed by long-term survival follow-up. **Onsite Visit Cadence:** Q3W during neoadjuvant phase (Weeks 0-6); Q6W during adjuvant phase (from Week 12). **Remote Monitoring Frequency:** Based on individual site and sponsor agreements. **Checklist Review Interval:** Ongoing at each dosing visit and post-surgery evaluations.

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					